



EMORY

Cell Tracking

Masoud Nateghi

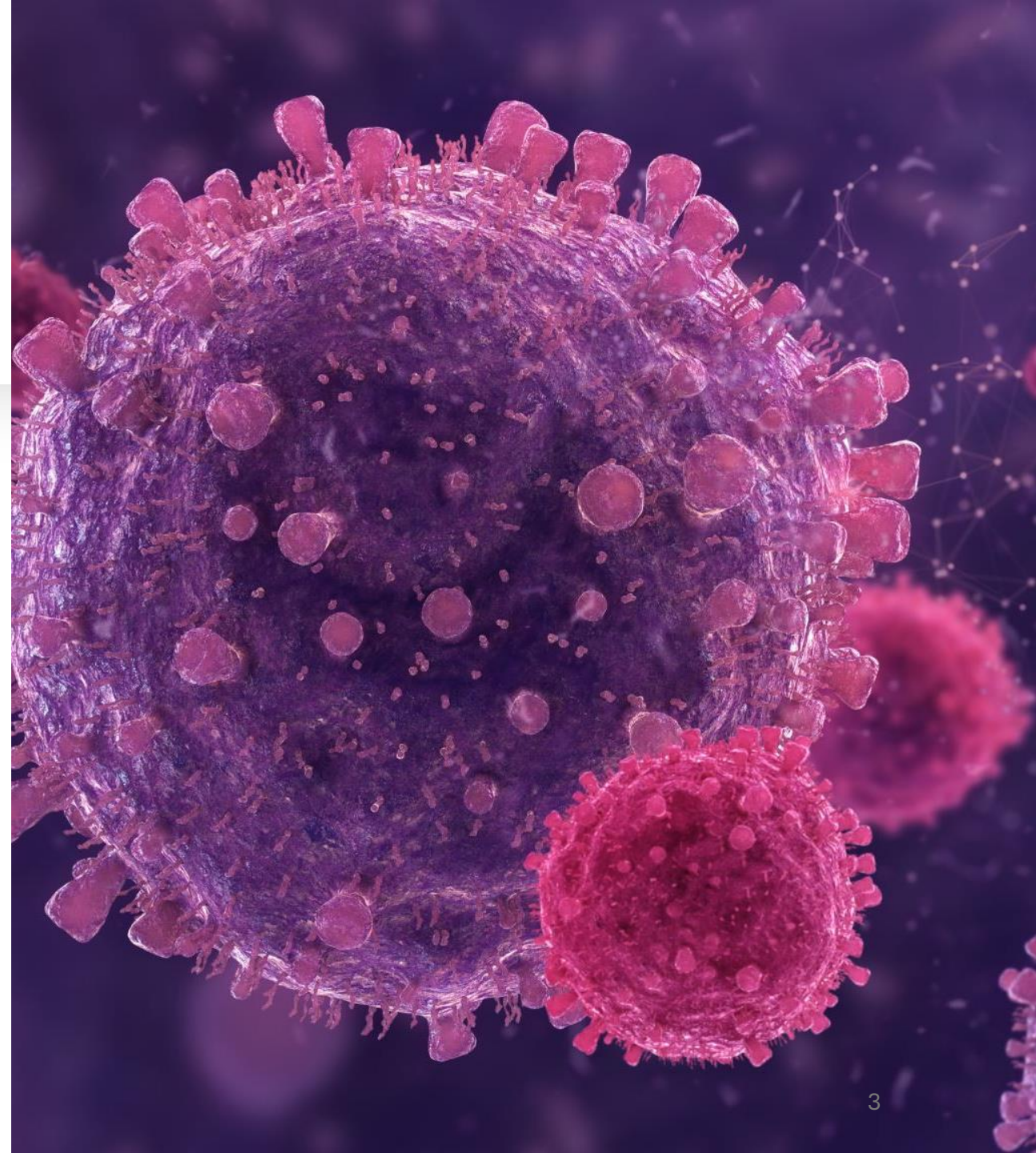
Introduction: A New Discovery in Cancer Research

- Traditional understanding vs. new findings
- How cancer cells spread through the body
- Impact on future cancer treatments



Introduction: The Problem: Cancer Metastasis

- **Cancer becomes deadly when it spreads (metastasis)**
- Cancer cells need to move through the body
- Understanding how they move is crucial for treatment



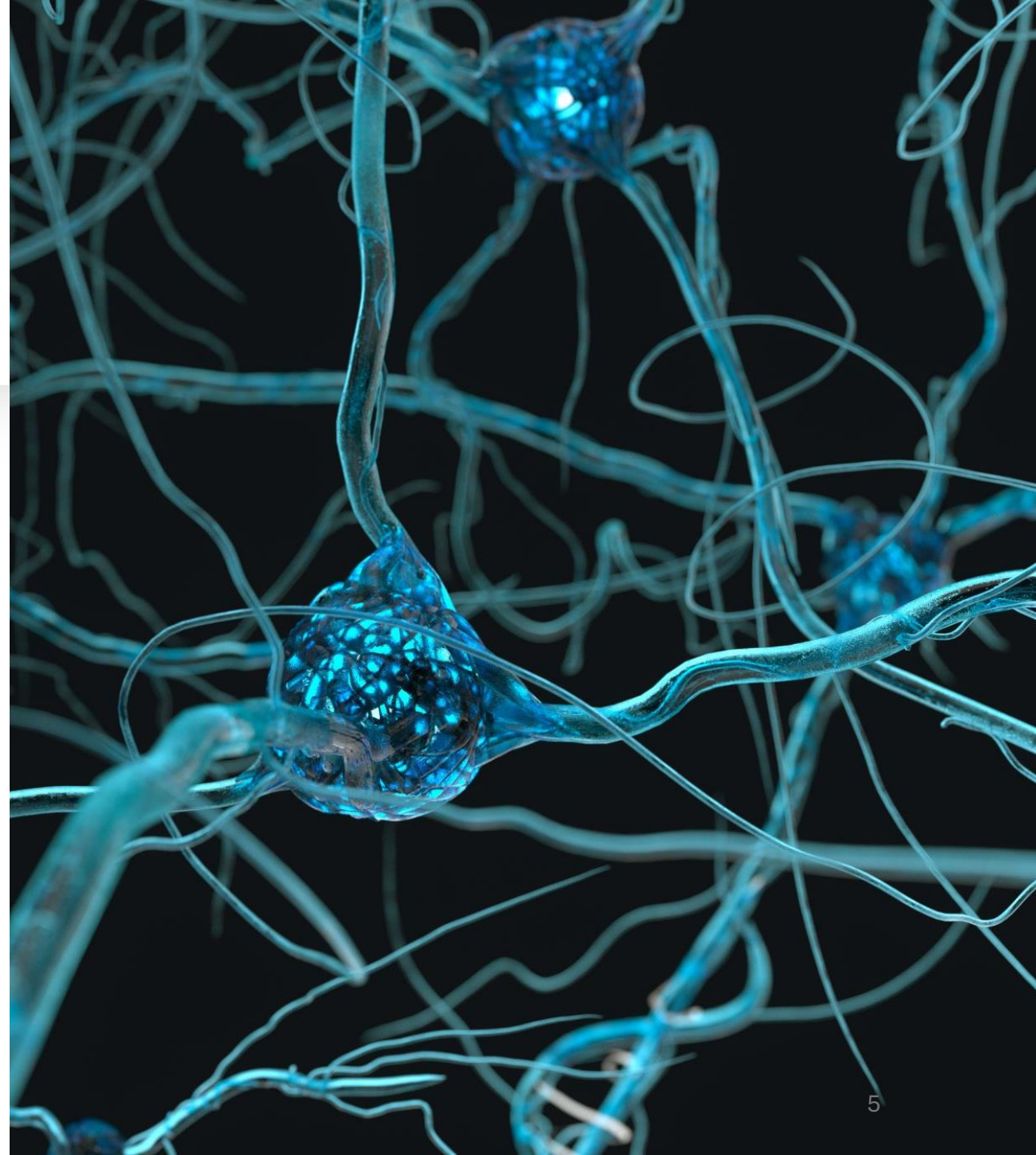
Introduction: Traditional Understanding of Cell Movement

- Cells mainly move using protein called myosin
- Acts like a cellular muscle
- Doesn't explain all cancer cell movement



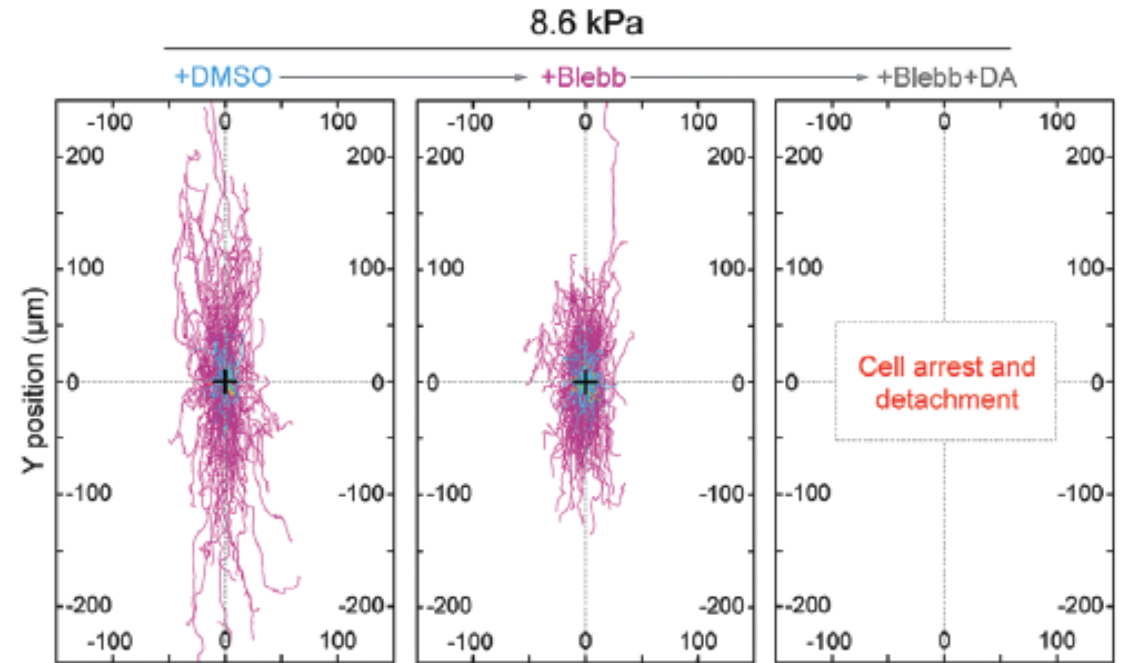
Introduction: New Discovery: The Dynein System

- Cancer cells use another protein: **dynein**
- Works like a **molecular motor**
- Pulls along microscopic fibers inside cells
- Provides **backup movement** system when myosin isn't working



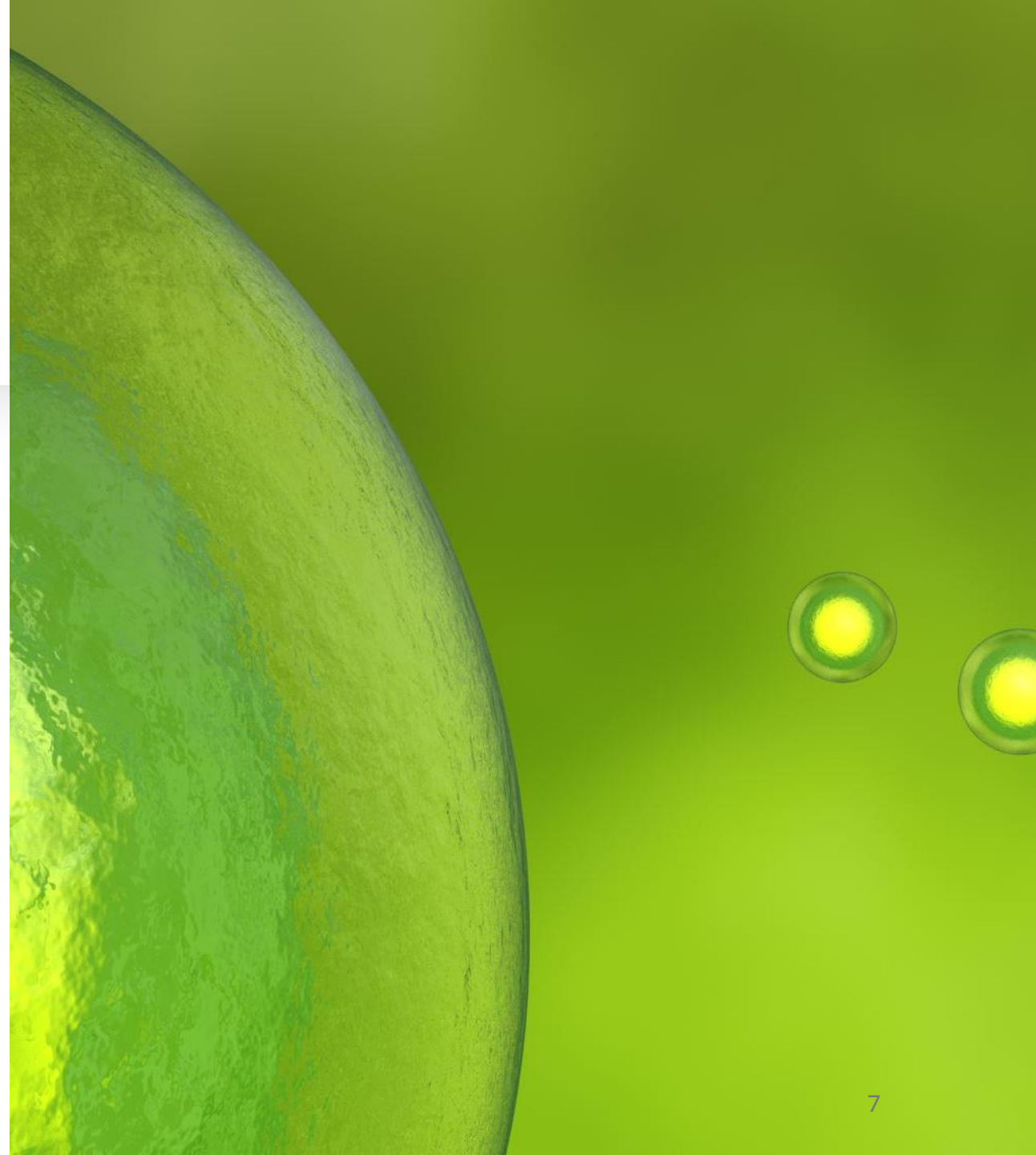
Introduction: What They Found

- Cells moved when myosin was blocked
- Cells moved when dynein was blocked
- Movement stopped only when both were blocked
- Confirms dual movement system exists
- A need for an algorithm which tracks cells



Materials: Dataset

- **Made a time-lapse video of cells movement**
- Kept cells alive at body temperature (37°C)
- **Took pictures every minute for several hours**
- Adding special **green dye** that doesn't harm them
- Using fluorescent tags that **light up** specific parts
- Taking both **regular and glowing pictures**

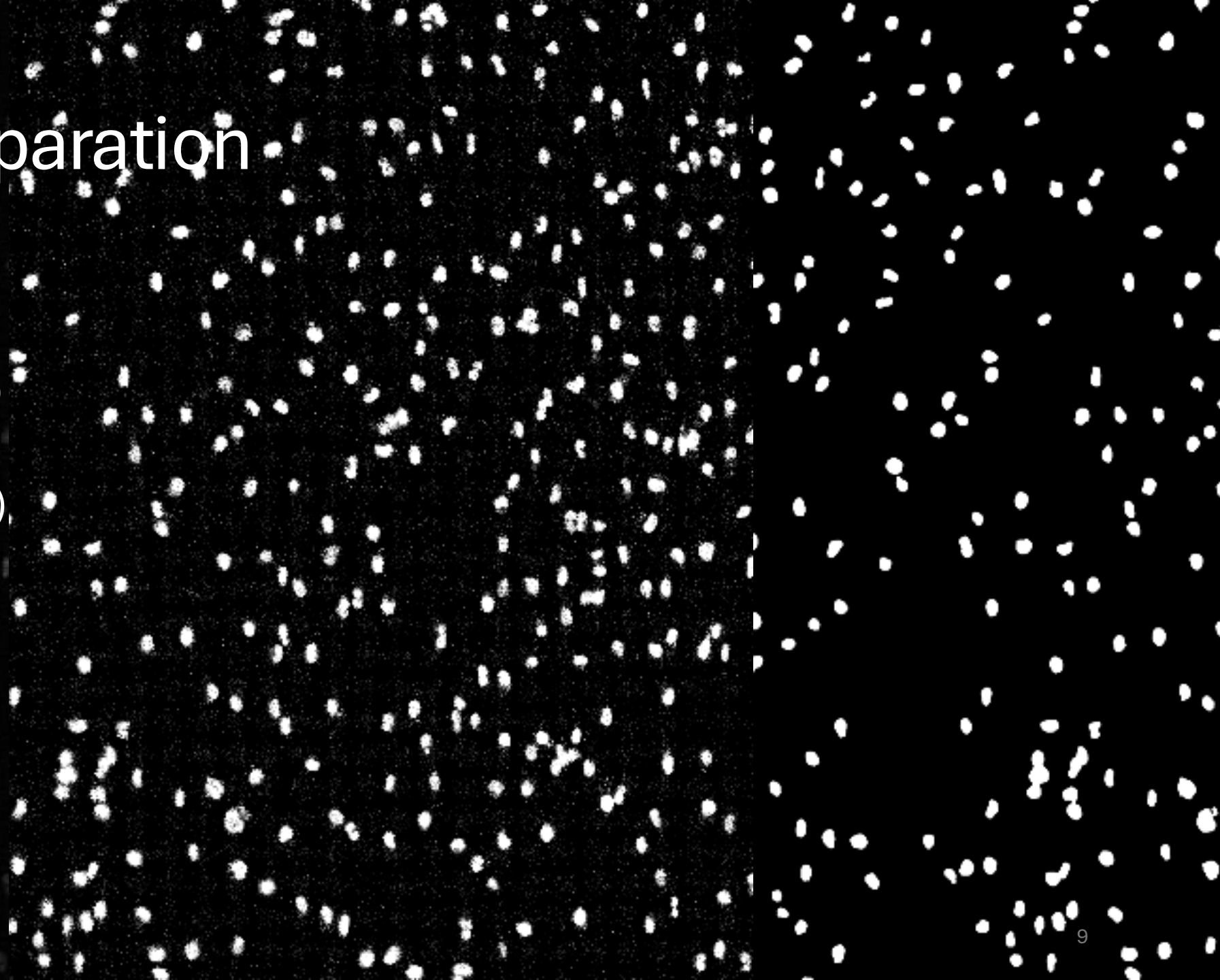


Materials: Dataset

- 2 channels of data of high-quality images (~2GB each)

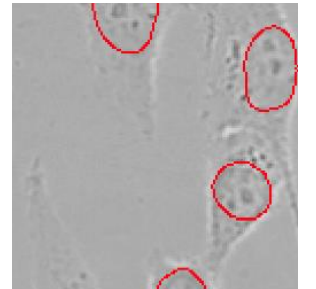
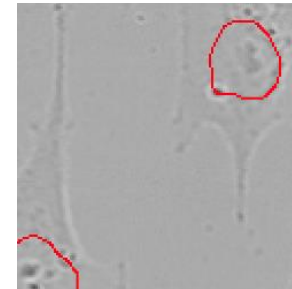
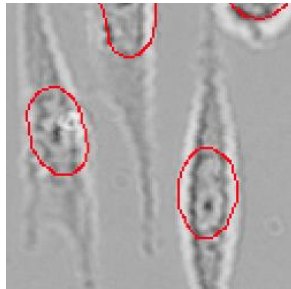
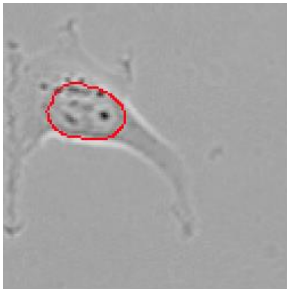
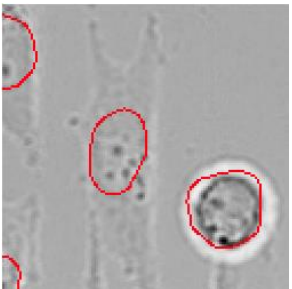
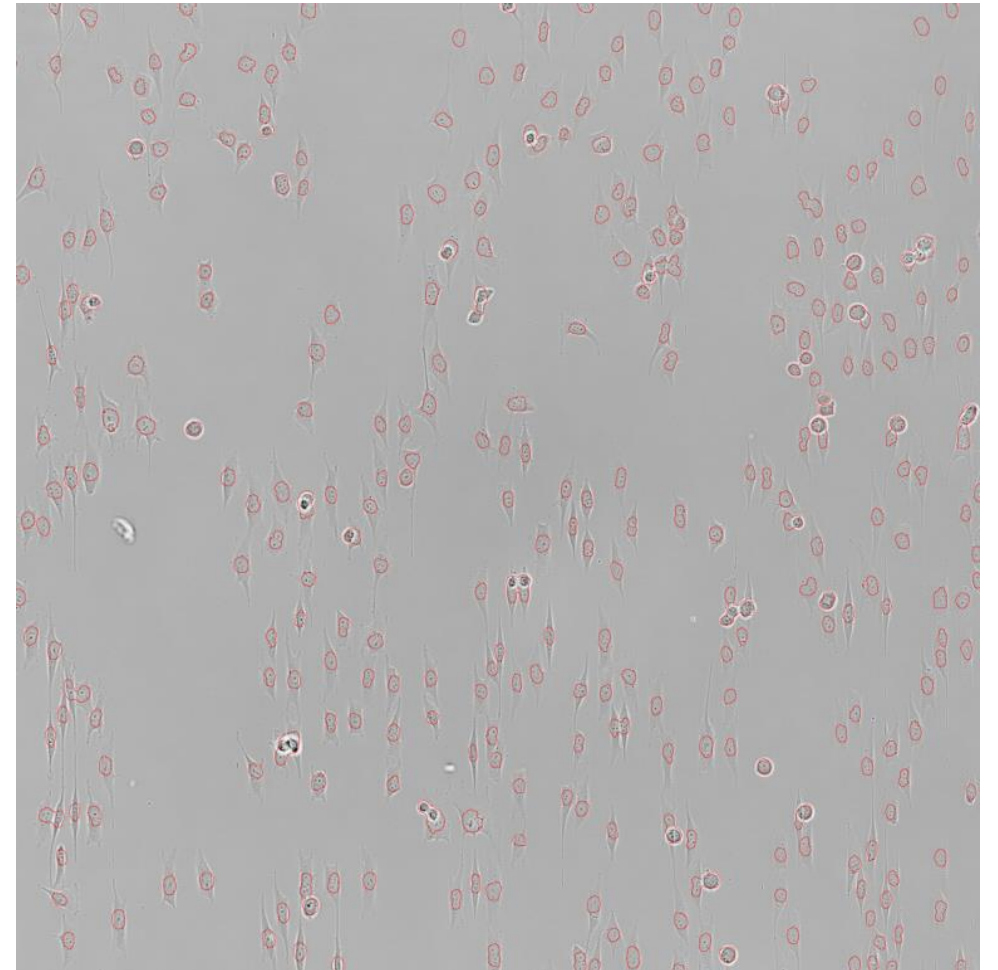
Dataset Preparation

- Mask:
 - Channel (2)
 - Resizing ($2048 * 2048$)
 - Thresholding (20)
 - Median Filtering ($9 * 9$)
 - Patch Size ($128 * 128$)

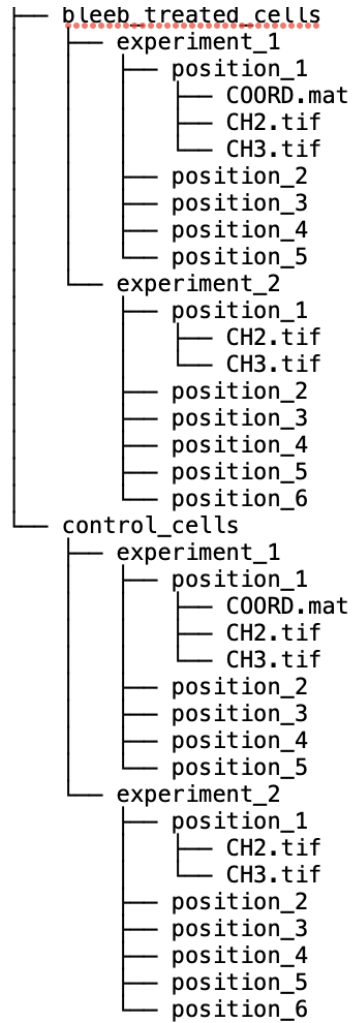


Dataset Preparation

- Image:
 - Resizing (2048 * 2048)
 - Thresholding (20)
 - Median Filtering (9 * 9)
 - Patch Size (128 * 128)
 - RED comes from channel 2.



Folder Structure



[↑](#) / home / mnategh / samenilab / team / masoud_nateghi / codes / cell-tracking / dataset /

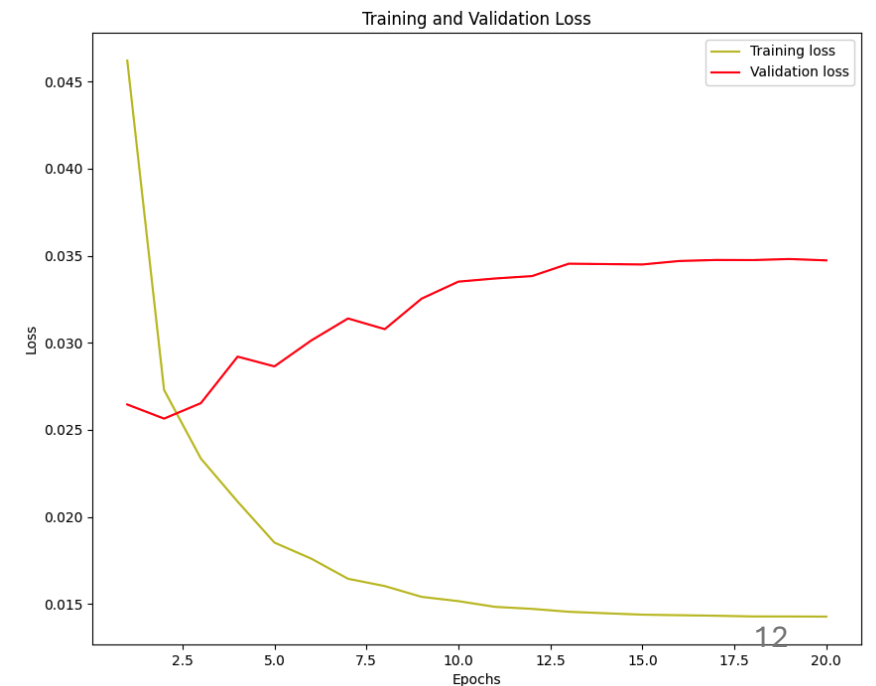
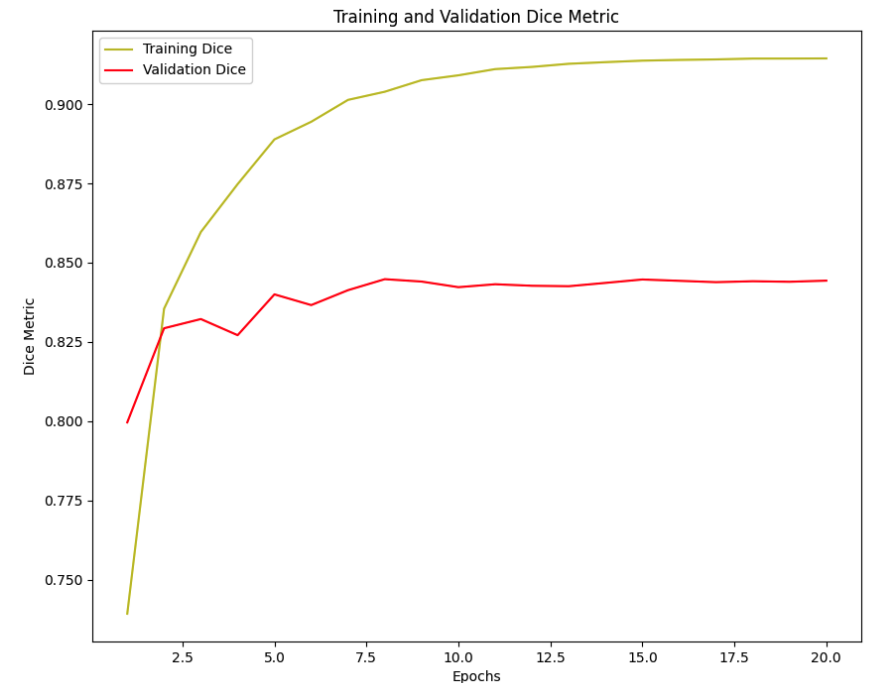
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☐	Folder	ce01p01	☐

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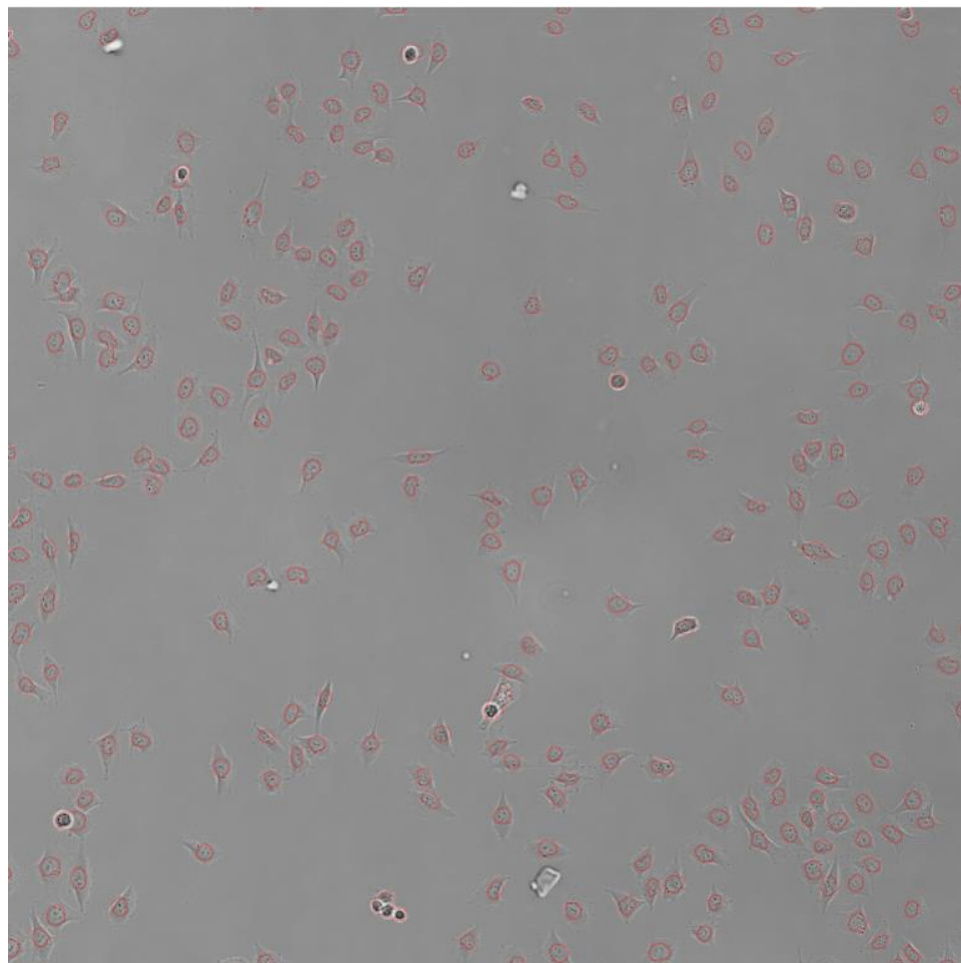
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Training Pipeline

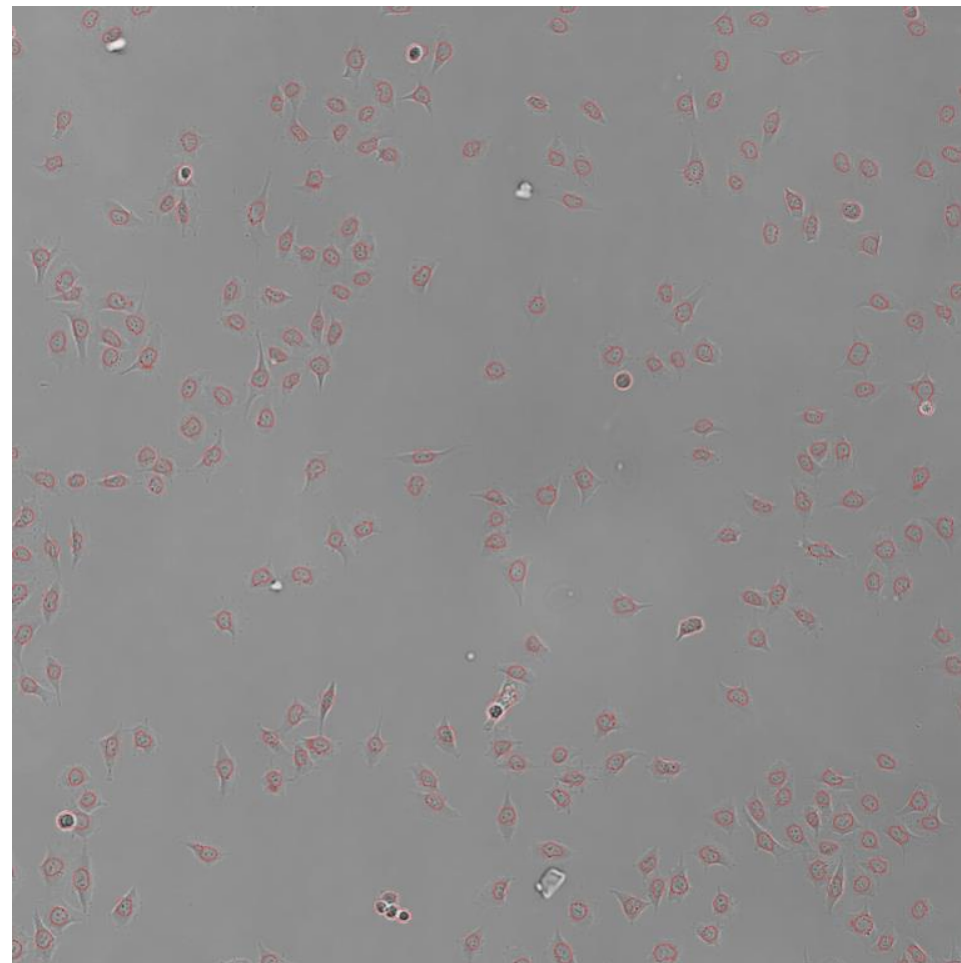
- Pre-trained U-net:
 - Backbone: Resnet34
 - Optimizer: Adam (lr=0.001)
 - Epochs: 20
 - Train set: 9 movies
 - Test set: 1 movie



Test (Unseen Data from Exp 1)

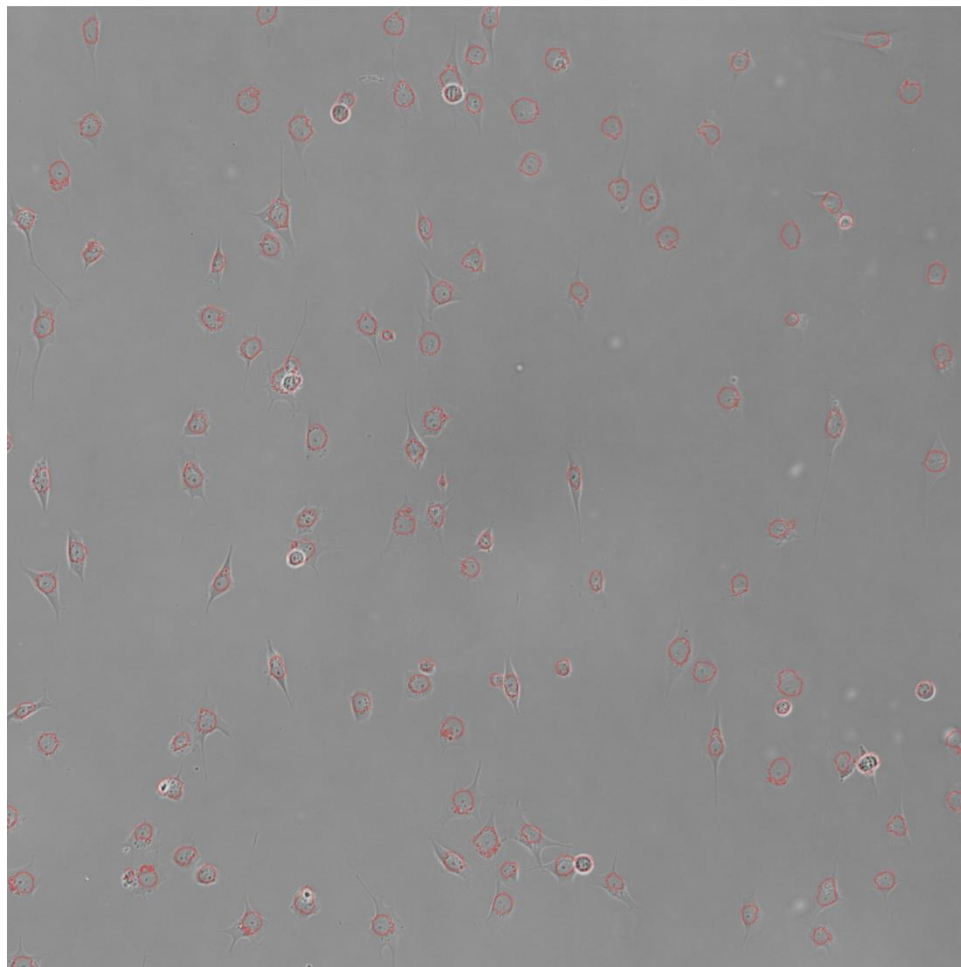


Ground Truth

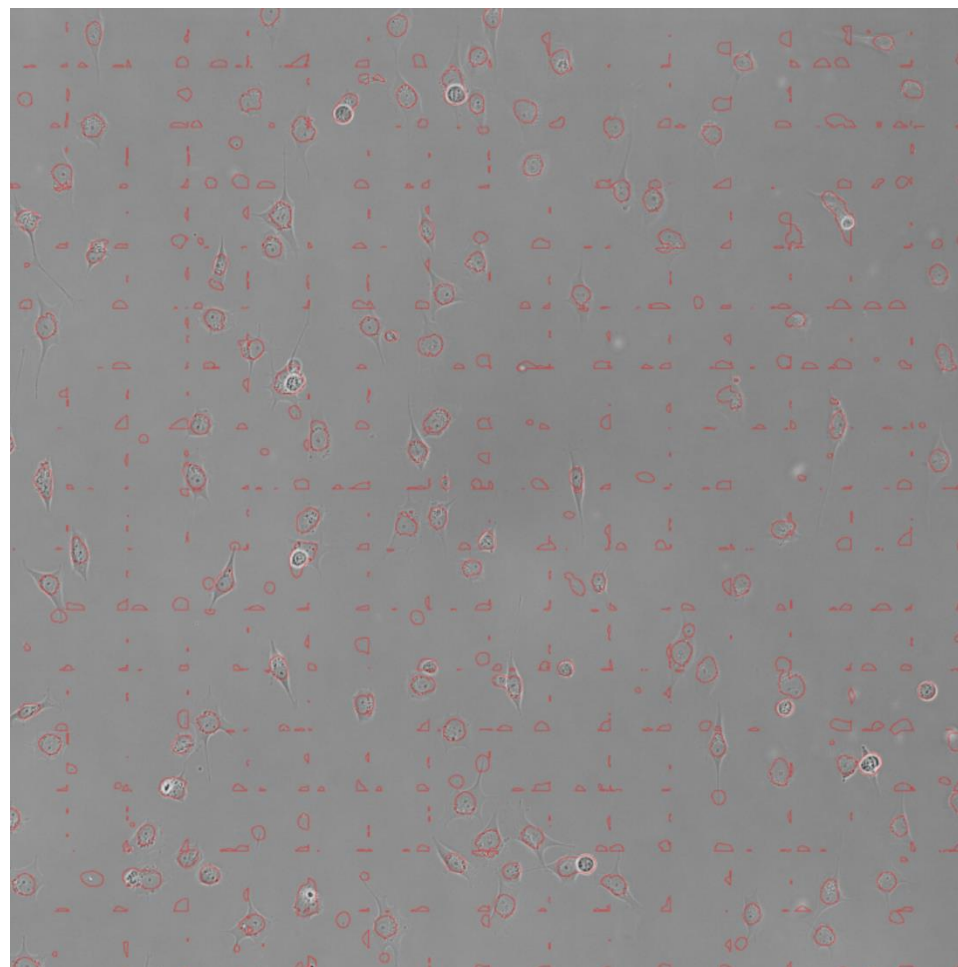


Predicted

Test (Unseen Data from Exp 2)



Ground Truth

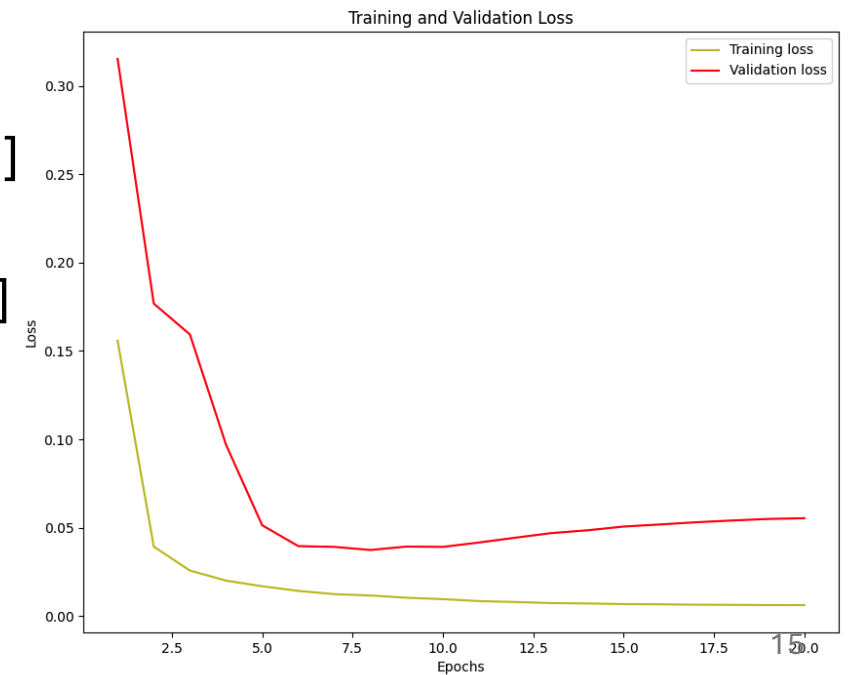
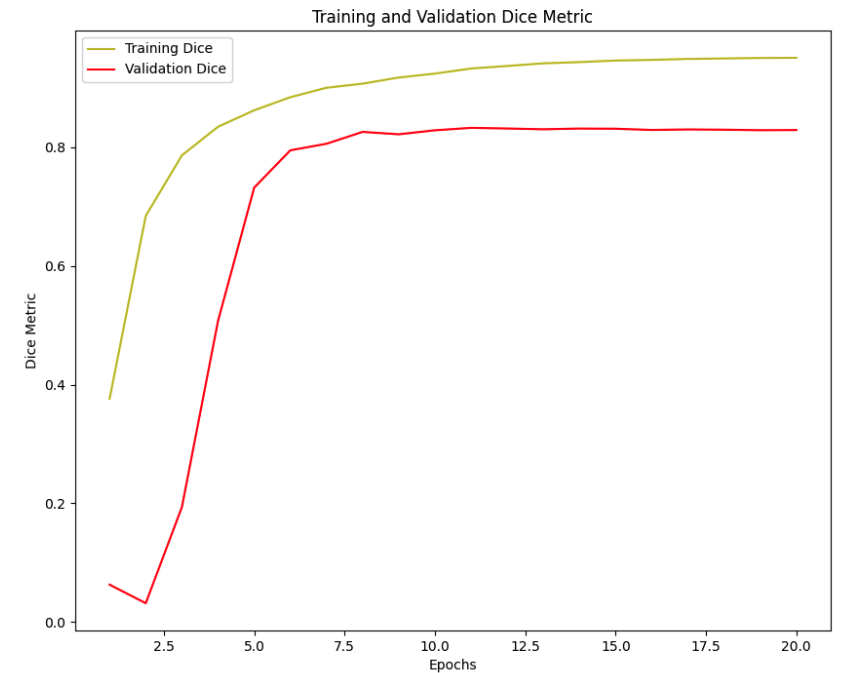


Predicted

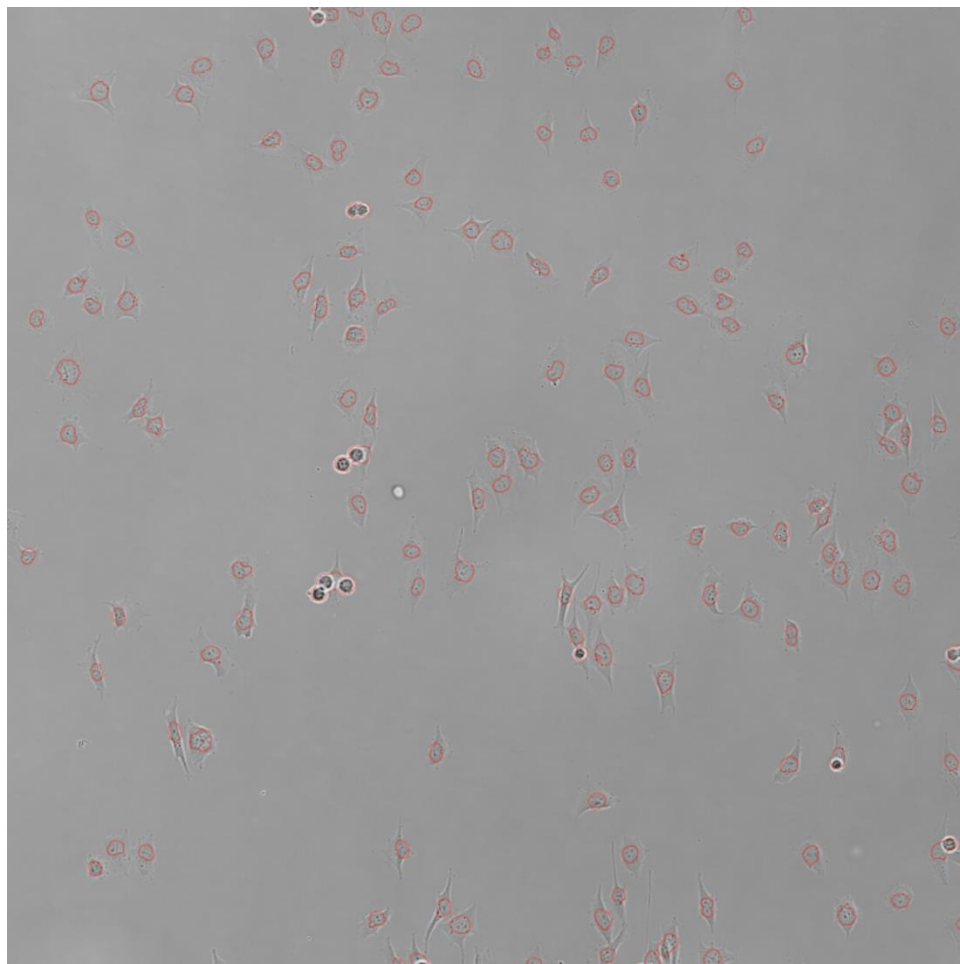
Training Pipeline

- Pre-trained U-net:
 - Backbone: Resnet34
 - Optimizer: Adam (lr=0.001)
 - Epochs: 20
 - Train set:
['be01p01','ce01p01',
'be02p01','ce02p01']

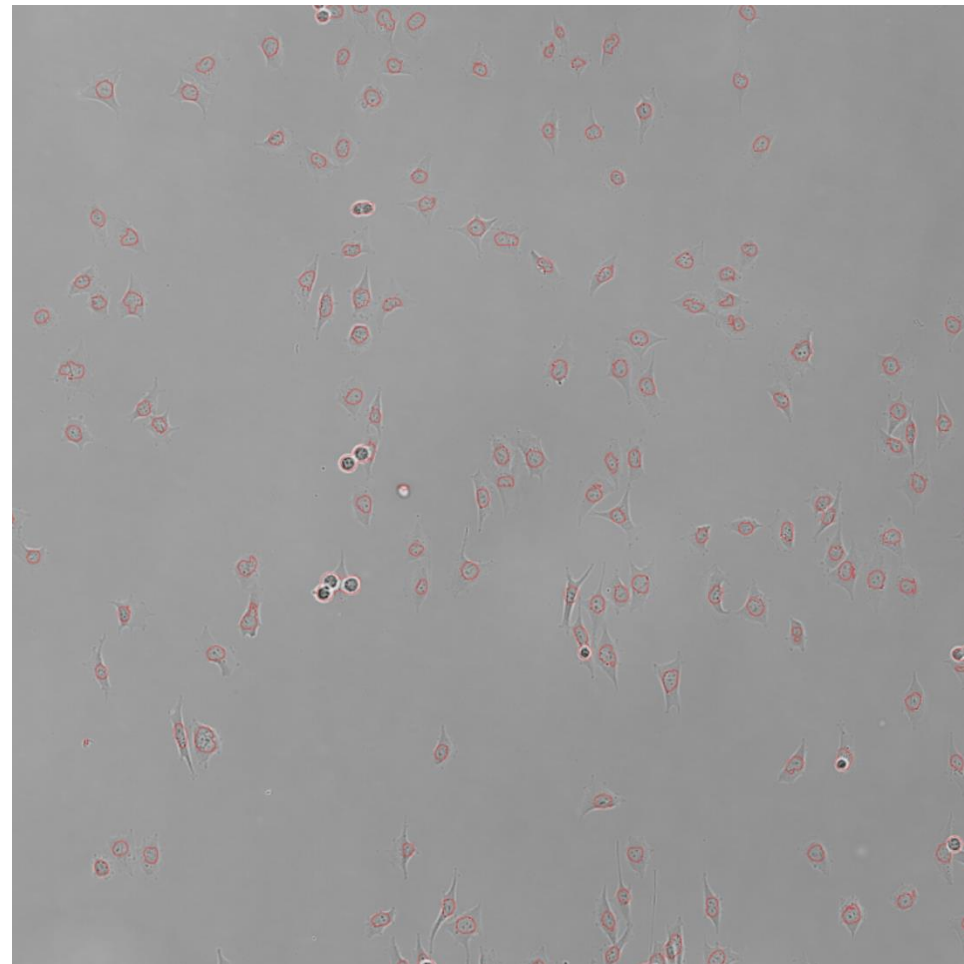
- Valid set:
['be02p03','ce01p05']



Test (Unseen Data from Exp 1)

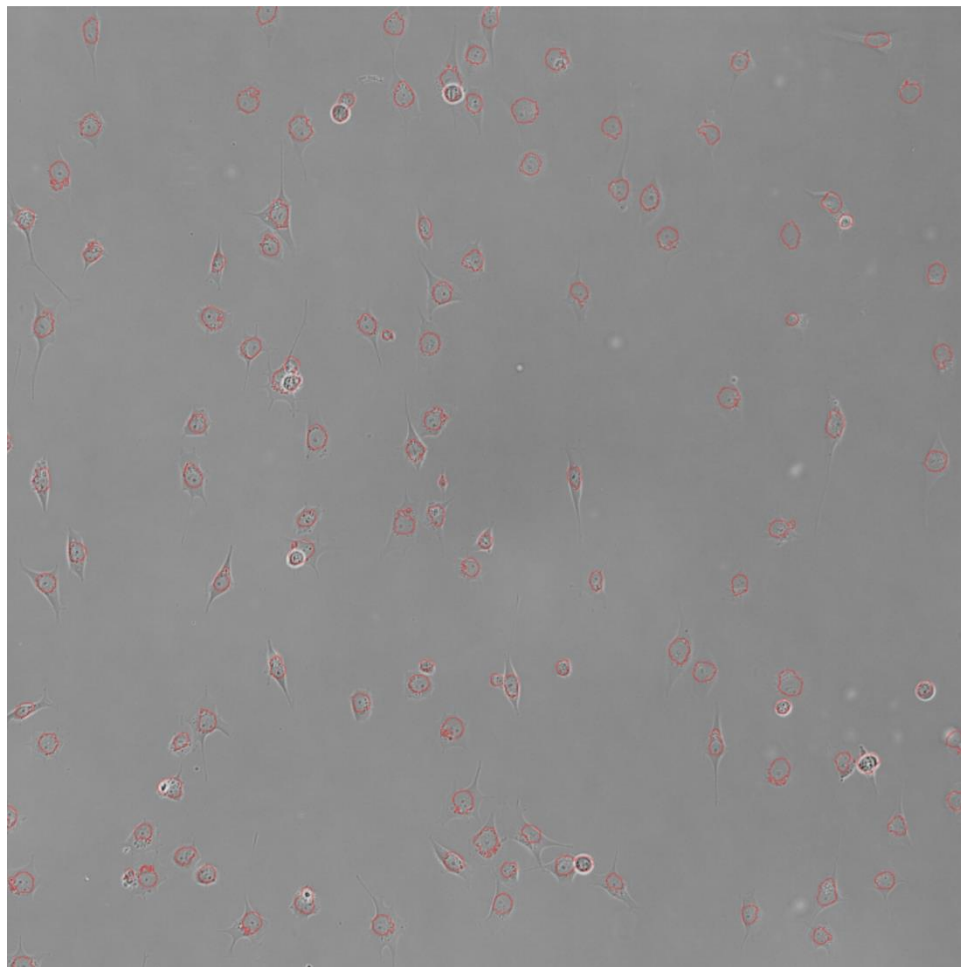


Ground Truth

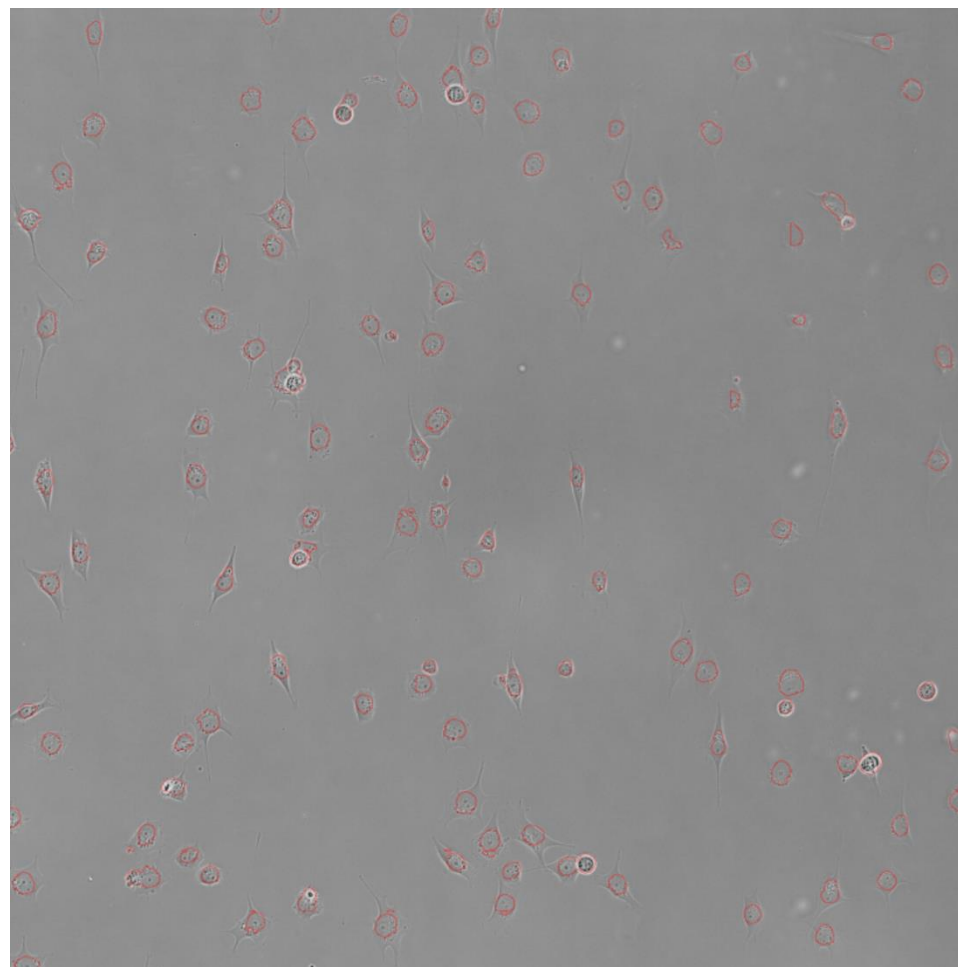


Predicted

Test (Unseen Data from Exp 2)



Ground Truth



Predicted

Future works

- Making a more robust and reliable ground truth (optical flow)
- Luckily, we have another channel identifying grids!
- Try other architectures as backbone of U-net
- Test-time augmentation for improving the accuracy
- YOLO

References

- Tagay Y, Kheirabadi S, Ataie Z, Singh RK, Prince O, Nguyen A, Zhovmer AS, Ma X, Sheikhi A, Tsygankov D, Tabdanov ED. Dynein-Powered Cell Locomotion Guides Metastasis of Breast Cancer. *Advanced Science*. 2023 Nov;10(31):2302229.
- Ronneberger O, Fischer P, Brox T. U-net: Convolutional networks for biomedical image segmentation. In *Medical image computing and computer-assisted intervention—MICCAI 2015: 18th international conference, Munich, Germany, October 5-9, 2015, proceedings, part III* 18 2015 (pp. 234-241). Springer International Publishing.



Questions